FloodRisk AI: A Physics-Informed Machine Learning System

Real-Time Flood Risk Assessment and Insurance Pricing

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Motivation & The Problem

The Global Protection Gap

- In 2023, natural catastrophe losses reached $280B USD; only ∼40% was insured.

FEMA NFIP Limitations:

1. **Resolution:** Zone-based lookups fail to capture property-scale topological variance.
2. **Stationarity:** Relies on historical boundaries, ignoring changing climate trends.

Our Solution: Replace static lookup tables with real-time, physics-based hydrological simulation accelerated by AI.

Mathematical Architecture

The system spans five interconnected mathematical frameworks:

System Component	Mathematical Tool
Rainfall Generation	Stochastic ODE, Poisson Process
Infiltration Model	System of First-Order ODEs
Spatial Flow Routing	10,000 Coupled Nonlinear ODEs
Neural ODE Acceleration	Adjoint Method, ODE Theory
Insurance Pricing	GEV Distribution, EAL Integral

Stochastic Rainfall Simulator

Rainfall $R(t)$ is modeled as a superposition of a seasonal baseline and a stochastic extreme-event process:

$$R(t) = R_{base} + A\sin(\omega t + \phi) + \sigma W(t) + \sum_{i=1}^{N(t)} R_i \cdot \delta(t - \tau_i)$$

- **Extreme Events:** $N(t) \sim \text{Poisson}(\lambda t)$ counts storm arrivals.
- **Intensity:** $R_i \sim \text{Pareto}(\alpha, x_m)$ captures the heavy right tail characteristic of extreme storms.

Infiltration: Anticipating the Stiffness Problem

Why an explicit solver (like RK4) fails here:

- The Green-Ampt equation $\frac{dF}{dt} = K_s \left(1 + \frac{\psi \Delta\theta}{F}\right)$ has a singularity at $F = 0$.
- This makes the ODE system highly *stiff*, requiring impractically small time steps to remain stable.

Our Method: Radau IIA

- We utilize a fully implicit Runge-Kutta scheme of order 5.
- It is unconditionally A-stable, allowing us to enforce mass conservation errors to $< 0.01\%$ via strict tolerances ($\epsilon_r = 10^{-6}$).

Spatial Flow Routing & Vectorization

We route water by solving 10,000 coupled nonlinear ODEs on a grid:

$$\frac{dh_i}{dt} = R_i(t) - I_i + \frac{1}{\Delta x} \sum_{j \in \mathcal{N}(i)} Q_{ji}(t)$$

How do we compute this fast enough?

- Iterating sequentially over 10,000 cells at every solver step is unfeasible.
- **Vectorization:** We pre-compute boundary index arrays ($Q_{ii} = 0$ for no-flow borders) and apply Manning's equation simultaneously via batched array operations.
- **Result:** RHS evaluation drops to $\sim$2 ms per step.

Neural ODE Acceleration & The Adjoint Method

The classical solver takes $\sim$8 seconds. We approximate it via a Neural ODE: $\frac{dz}{dt} = f_\theta(z, t)$.

How do we train this without memory explosion?

- Storing intermediate integration steps for backpropagation requires $\mathcal{O}(T)$ memory.
- **Solution:** The Adjoint Sensitivity Method computes gradients by solving a second ODE backwards in time:

$$\frac{da}{dt} = -a(t)^T \frac{\partial f_\theta}{\partial z}, \quad \frac{d\mathcal{L}}{d\theta} = -\int_T^0 a(t)^T \frac{\partial f_\theta}{\partial \theta} dt$$

- Yields exact gradients with $\mathcal{O}(1)$ memory. Inference drops to **287 ms**.

Risk Quantification

Extreme Value Theory: To capture tail risk, we fit a Generalised Extreme Value (GEV) distribution (Fréchet family, $\xi > 0$) to annual max depths.

Actuarial Insurance Pricing:

- Expected Annual Loss (EAL): $\text{EAL} = V \int_0^\infty D(h)p(h)dh$
- Gross Premium (with risk loading $\theta = 0.30$ and expense ratio $e = 0.28$):

$$\text{Premium} = \frac{\text{EAL} \cdot (1 + \theta)}{1 - e}$$

Results & Validation

Testing the Physics Engine against Actuarial Data:

- Test Subject: Residential property valued at $450,000.
- Physics-based pure premium computed: **$3,897**.
- Historical FEMA NFIP benchmark: **$3,690**.
- **Variance of** $+5.6\%$ — strong validation considering our model uses real-time differential equations rather than decades of historical claims.

Robustness Checks: Physical Invariants

Common Question: How do you guarantee the AI obeys physical laws?

- **Non-negative Depths:** The Neural ODE Decoder uses a *Softplus* activation ($\log(1 + e^x)$). Unlike ReLU, it is differentiable everywhere (crucial for the Dormand-Prince solver) and strictly guarantees $h \geq 0$.
- **Physics-Informed Loss Function:**

$$\mathcal{L} = \text{MSE} + \lambda_1 \sum \max(0, -\hat{H}_i)^2 + \lambda_2(\|\nabla_x \hat{H}\|^2 + \|\nabla_y \hat{H}\|^2)$$

- **Mass Conservation verified via 45 automated tests:** The mathematical engine enforces relative error $\epsilon_{rel} < 0.01\%$ at every step.

Limitations & Future Scalability

Common Question: How does this scale to complex, real-world topography?

- **Current Limitation:** The model currently uses synthetic terrain.
- **Future Integration:** The system is built to ingest USGS 3DEP 1-meter elevation data.
- **Hydraulic Fidelity:** Manning's equation assumes a shallow water approximation. To scale to highly complex terrain, future iterations will replace Manning's flow with the full **2D Saint-Venant equations**.
- **Temporal Clustering:** We will incorporate the *extremal index* to account for storms that happen in clusters, rather than assuming independence of annual maxima.

Thank You